Fire-Sale Cascade Simulator · One-page spec

Case 02 · A Slow-Motion Crisis · Stage 04: Specification

The problem

In the UK in 2022 and at SVB in 2023, the shock did not break the system; the chain reaction did. One forced sale pushed the price down, and the next institution had to sell. Standard stress tests look at each institution alone, and no open tool shows the market as one system.

The promise

A simulation that shows, round by round, how one sale triggers the next; that never gives a single number but a range and what drives it; and that runs entirely on the user's machine, so no balance sheet ever leaves it.

Ten parts in four layers

- **World:** 1 · World builder with five agent types and signed official worlds. 2 · Balance sheets and valuation, with accounting vs economic capital. 3 · Behavioral rules as readable text.
- **Cascade:** 4 · Three price-impact models, with cross-impact. 5 · Round engine with conservation checks. 6 · Lending network, direct contagion and DebtRank.
- **Questions:** 7 · Four real-world interventions, with triggers. 8 · Uncertainty sweep, sensitivity analysis and tipping point.
- **Trust:** 9 · Calibration against the UK 2022 and SVB 2023, with a public fit report. 10 · Cascade view and reproducible links, all in the browser.

Six decisions

- Five agent types, because the UK and SVB were two different mechanisms.
- Rules as validated text, because they are among the model's biggest limits.
- A range, not a number, because the answer flips within one point of leverage.
- Browser-only execution, so real balance sheets never reach a server.
- Three visible price-impact models, because this is the most sensitive part.
- A fit report that shows failures too, in red.

Still open

- The result depends mostly on exposure data nobody sees. The tool can say how much, and calibrate on past crises.
- For today's market it still answers "it depends on leverage we don't know".